

Chapter 4: In-Class Exercises

4.2

Problem: Chinh is presented an annuity that is to pay \$4,000 at the end of each six months for six years. Using compound interest at a nominal monthly interest rate of 3%, what is the value of Chinh's reward at the time of the announcement (six months before the first \$4,000 payment)?

4.2

Problem: Josh makes annual beginning-of-year contributions of \$3,000 to a retirement account. The account has a nominal interest rate of 4.8% convertible monthly. Find Josh's balance just before his ninth contribution by finding the annual effective interest rate and then check your answer using (4.2.8).

4.2

Problem: A perpetuity paying \$1,650 at the end of each year is to be replaced by another paying X at the beginning of each two years. The present value of each perpetuity is \$31,250. Find X .

4.3

Problem: Mahpee was offered a scholarship with level monthly payments for four years. The first payment would be made on September 30, 2000, and each year the payments would total \$5,904. Mahpee, who was contemplating actuarial studies, wished to compare this scholarship to a fellowship offered by another college. The latter would pay \$5,800 on September 1st of years 2000, 2001, 2002, and 2003. Assume that the growth of money is governed by compound interest at an annual effective interest rate of 4%, and compare the values of the two grants on September 1, 2000.

4.3

Problem: Shirley invests \$200 at the end of each quarter for eight years in an account having an annual effective interest rate of 4%. Find Shirley's accumulated value at the time of her last deposit.

4.3

Problem: Find the numerical values of the four annuity symbols in inequality (4.3.14) when $n = 48$, $m = 12$, and $i = 5\%$.

4.3

Problem: Pavol purchases a perpetuity-due. The perpetuity has monthly payments of \$100 during the first five years, and then these increase to \$300. Determine the purchase price if it is based on an annual effective discount rate of 3.6%.

4.6

Problem: Martin deposits \$12,000 in a savings account with a discount rate of 10% convertible quarterly. He leaves his money in this account to accumulate for fifteen years, then moves it to a fund that is accumulating at 8% per annum convertible continuously. If, starting at time 15 when he invests in the new fund, money is withdrawn continuously at a rate of \$6,000 per annum, how long will Martin's money last?

4.6 **Problem:** Suppose that money grows according to simple discount at a rate of 3.6%. Find formulas for $\bar{a}_{\overline{n}|}$, $\bar{s}_{\overline{n}|}$, $\bar{a}_{\overline{n}|}$.

4.6 **Problem:** An annuity is payable continuously for ten years. The rate of payment at time t is t^2 yearly. Assuming there is a constant force of interest $\delta = .04$, find the present value of the annuity.